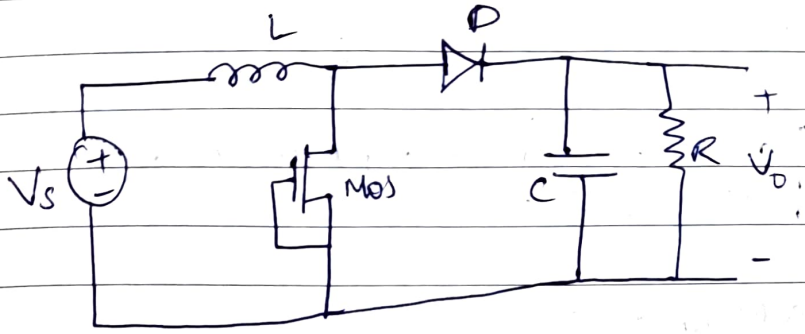
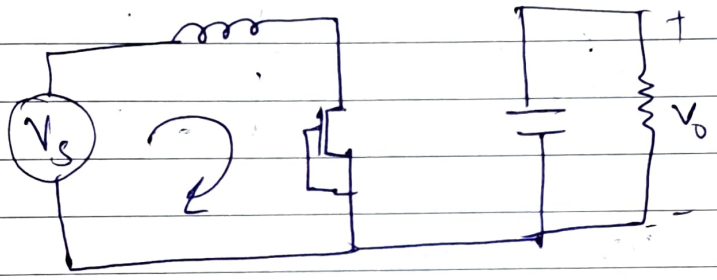
Boost Converter



Mode 1 - (MOSFET ON)

Duration → $DT_s > t > 0$



KVL

$$V_{in} - V_L = 0$$

$$\boxed{V_{in} = V_L} \rightarrow \text{inductor voltage}$$

$$V = L\frac{di}{dt}$$

$$V_{in} = L\frac{di_L}{dt}$$

$$\boxed{\frac{V_{in}}{L} = \frac{di_L}{dt}} \rightarrow \text{inductor current}$$

⇒) current entering capacitor = 0

→) current leaving capacitor = load current

$$I_o = \frac{V_o}{R}$$

$$I_C = -I_o$$

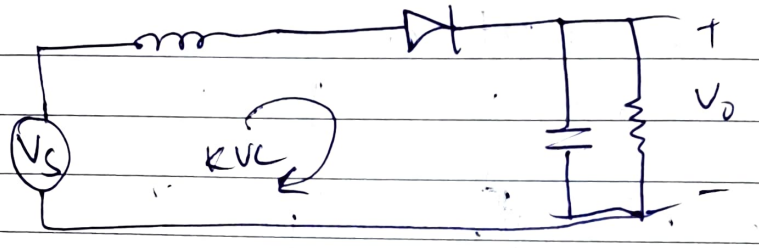
$$\boxed{I_C = -\frac{V_o}{R}} \rightarrow \text{capacitor current.}$$

$$I = C\frac{dv}{dt}$$

$$\frac{-V_o}{R} = C\frac{dV_o}{dt} \qquad (V_o = V_c)$$

$$\boxed{\frac{dV_o}{dt} = \frac{-V_o}{RC}} \rightarrow \text{capacitor voltage}$$

MODE-2 ⇒ Mosfet off



$$V_{in} - V_L - V_o = 0$$

$$\boxed{V_L = V_{in} - V_o}$$

$$V = L\frac{di}{dt}$$

$$\boxed{\frac{di_L}{dt} = \frac{V_{in} - V_o}{L}} \rightarrow \text{inductor current.}$$

KCL: $I_c = I_L - I_o$

$$I_o = \frac{V_o}{R}$$

$$\boxed{I_c = I_L - \frac{V_o}{R}} \rightarrow \text{capacitor current}$$

$$I = C\frac{dv}{dt} \Rightarrow I_L - \frac{V_o}{R} = C\frac{dV_o}{dt}$$

$$\boxed{\frac{dV_o}{dt} = \frac{I_L - \frac{V_o}{R}}{C}} \rightarrow \text{capacitor voltage}$$

Volt Sec law for boost converter

$$D V_{L(on)} + (1-D) V_{L(off)} = 0$$

$V_L = V_{in} \rightarrow$ on

$V_L = V_{in} - V_o \rightarrow$ off

$$DV_{in} + (1-D)(V_{in} - V_o) = 0$$

$$DV_{in} + V_{in} - DV_{in} - V_o + DV_o = 0$$

$$DV_{in} - DV_{in} = 0$$

$$V_{in} - V_o + DV_o = 0$$

$$V_{in} - V_o(1-D) = 0$$

$$V_o(1-D) = V_{in}$$

$$\boxed{V_o = \frac{V_{in}}{1-D}}$$

$$\boxed{D = 1 - \frac{V_{in}}{V_o}}$$

Volt sec Balance law

$$V_L = L \frac{di}{dt}$$

$$\int_0^{T_s} V_L dt = L \int_0^{T_s} \frac{di_L}{dt} dt$$

$$\boxed{\int_0^{T_s} V_L dt = L\,(i_L(T_s) - i_L(0))}$$

In steady state, $i_L(T_s) = i_L(0)$

$$\therefore \boxed{\int_0^{T_s} V_L dt = 0} \Rightarrow \frac{1}{T_s} \int_0^{T_s} V_L dt = 0$$

$$\boxed{\langle V_L \rangle = 0}$$

Ampere second balance

$$I_c = C\frac{dV_c}{dt}$$

$$\int_0^{T_s} I_c\,dt = C\int_0^{T_s}\frac{dV_c}{dt}\,dt$$

$$C[V_c(T_s) - V_c(0)]$$

$$\int_0^{T_s} I_c\,dt = C[V_c(T_s) - V_c(0)]$$

Steady state → $V_c(T_s) = V_c(0)$

$$\therefore \boxed{\int_0^{T_s} I_c\,dt = 0} \rightarrow \frac{1}{T_s}\int_0^{T_s} I_c\,dt = 0$$

$$\boxed{\langle I_c \rangle = 0}$$

Ampere sec on boost converter

$$I_c = \frac{V_o}{R} \rightarrow \text{on}$$

$$I_c = I_L - \frac{V_o}{R} \rightarrow \text{off}$$

$$D\left(-\frac{V_o}{R}\right) + (1-D)\left(I_L - \frac{V_o}{R}\right) = 0$$

$$-\frac{DV_o}{R} + (1-D)I_L - \frac{(1-D)V_o}{R} = 0$$

$$D + (1-D) = 1$$

$$-\frac{V_o}{R} + (1-D)I_L = 0$$

$$\boxed{I_L = \frac{V_o}{R(1-D)}} \Rightarrow \boxed{I_L = \frac{V_{in}}{R(1-D)^2}}$$

Inductor Ripple

on :- $\frac{di}{dt} = \frac{V_{in}}{L}$

ΔI_L = slope x time $\Rightarrow \Delta I_L = \frac{V_{in} D T_s}{L}$

$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

Inductor Design

$$L = \frac{V_{in} D}{\Delta I_L f_s}$$

Capacitor Ripple

$$I_C = -I_o$$

$$I_C = C \frac{dv}{dt}$$

$$\Delta V_o = \frac{I_o D T_s}{C}$$

$$\boxed{\Delta V_o = \frac{V_o D}{R C f_s}}$$

Parameters derive

$V_{in} = 12V$

$V_o = 20V$

Duty cycle → $V_o = \frac{V_{in}}{1-D}$

$$20 = \frac{12}{1-D}$$

$$\boxed{D = 0.4 \text{ or } 40\%}$$

taking $R = 4\Omega$

Output Voltage → $\boxed{I_o = \frac{V_o}{R} = \frac{20}{4} = 5A}$

Avg Inductor current → $I_L = \frac{I_o}{1-D}$

$$I_L = \frac{5}{0.6}$$

$$\boxed{I_L = 8.33A}$$

Inductor ripple → $\Delta I_L = \frac{V_{in}D}{Lf_s}$

$$L = \frac{V_{in}D}{\Delta I_L f_s}$$

$D = \frac{T_{on}}{T_s}$ → switching period

$0.4 \times T_s = T_{on}$

ripple current → $\boxed{30\% \times I_L = 2.5A}$

taking $T_s = 40\mu s$

$$f_s = \frac{1}{40\times10^{-6}} = 25KHz$$

Inductor Value (L)

$$L = \frac{V_{in}D}{\Delta I_L f_s}$$

$$L = \frac{12\times0.4}{2.5\times25000}$$

$$L = \frac{12\times0.4}{62500}$$

$$L = 76.8\times10^{-6}H$$

Capacitor Value (C)

$$\Delta V_o = \frac{I_o D}{Cf_s}$$

$$C = \frac{I_o D}{\Delta V_o f_s}$$

$\Delta V_o \rightarrow 0.2$ (voltage dip ripple)

$$C = \frac{5\times0.4}{0.2\times25000}$$

$$C = 400\mu f$$

output power

$$P_o = V_o I_o$$

$$= 20\times5$$

$$P_o = 100W$$

(ripple current & ripple voltage cannot be derived
30% 0.2 have to assume)

~~State space Representation~~
state space Averaging

→ Aboost convetor is not a single linear circuit
→ It has 2modes 1) on 2) off

→ The system keeps switching between these two every switching period.
→ classical control theory (transfer functions, Bode plot, PID controllers) requires one linear model
→ we need to combine these two models into one equivalent model
→ That process is called state space Averaging

Step 1 $x = \begin{bmatrix} i_L \\ V_c \end{bmatrix}$ (i_L: inductor current; V_c: output voltage)

I/p : $V_{in} = u$

O/p : $V_c = V_o = y$

Step 2 Onstate eqns :- $\frac{di_L}{dt} = \frac{V_{in}}{L}$ —①

$\frac{dV_c}{dt} = -\frac{V_c}{RC}$ —②

state eqn :- $\dot{x} = A_1 x + B_1 u$

$\frac{di_L}{dt} = (0)x(i_L) + 0 \times V_c + \frac{1}{L} V_{in}$ —①

$\frac{dV_c}{dt} = 0 \times i_L - \frac{1}{RC} V_c$ —②

$A_{on} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix}$; $B_{on} = \begin{bmatrix} 1/L \\ 0 \end{bmatrix}$

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Step 3 → Off state eqns :-

$$\frac{di_L}{dt} = \frac{V_{in} - V_c}{L} \quad -①$$

expanding $\Rightarrow 0 \times V_{in} + 0 \times i_L - \frac{1}{L} V_c + \frac{1}{L} V_{in}$

$$\frac{dV_c}{dt} = \frac{i_L}{C} - \frac{V_c}{RC} \quad -②$$

$$A_{off} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix} \; ; \; B_{off} = \begin{bmatrix} 1/L \\ 0 \end{bmatrix}$$

Step 4 → Averaging :-

(D : 40%)

$$A = D A_{on} + (1-D) A_{off}$$

$$B = D B_{on} + (1-D) B_{off}$$

Average A matrix → $A_{on} = \begin{bmatrix} 0 & 0 \\ 0 & -1/RC \end{bmatrix} \; ; \; A_{off} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix}$

$$A = D \begin{bmatrix} 0 & 0 \\ 0 & -1/RC \end{bmatrix} + (1-D) \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -\frac{(1-D)}{L} \\ \frac{1-D}{C} & -1/RC \end{bmatrix}$$

$$B = B_{on} = B_{off} = \begin{bmatrix} 1/L \\ 0 \end{bmatrix}$$

$$B = D \begin{bmatrix} 1/L \\ 0 \end{bmatrix} + (1-D) \begin{bmatrix} 1/L \\ 0 \end{bmatrix} = \begin{bmatrix} 1/L \\ 0 \end{bmatrix}$$

$$\frac{d(V-\hat{v})}{dt} = \frac{1-(D+\hat{d})}{C} I_L + \hat{i}_L - \frac{V_c + \hat{v}_c}{RC}$$

Step 7 → Expand the product

the nonlinear term becomes $(1-D-\hat{d})(V+\hat{v})$

$$= (1-D)V + (1-D)\hat{v} - V\hat{d} - \hat{d}\hat{v}$$ ($\hat{d}\hat{v} \to 0$)

$\hat{d}\hat{v}$ → is extremely small → ignore

$(1-D)V$ → DC terms

$(1-D)\hat{v}$, $-V\hat{d}$, $\hat{v}_g$ → AC terms

(allow all AC terms)

Step 8 → Final small signal equations

Inductor eqn → $$\frac{d\hat{i}_L}{dt} = -\left(\frac{1-D}{L}\right)\hat{v} + \frac{V}{L}\hat{d} + \frac{1}{L}\hat{v}_g$$

Capacitor eqn → $$\frac{d\hat{v}}{dt} = \frac{1-D}{C}\hat{i}_L - \frac{I_L}{C}\hat{d} - \frac{\hat{v}}{RC}$$

0 if Ip voltage is const

Derivation of Transfer function

$G_{vd}(s)$ = gain = $\frac{\text{output Voltage}}{\text{Duty cycle}}$

→ in small signal form, $\boxed{G_{vd}(s) = \frac{\hat{v}(s)}{\hat{d}(s)}}$

Step 1:- input voltage disturbance is zero

$\boxed{\hat{V}_g = 0}$

∴ The eqns become

$$\frac{d\hat{i}_L}{dt} = -\frac{1-D}{L}\hat{v} + \frac{V}{L}\hat{d}$$

$$\frac{d\hat{v}}{dt} = \frac{1-D}{C}\hat{i}_L - \frac{I_L}{C}\hat{d} - \frac{\hat{v}}{RC}$$

Step 2:- Apply Laplace Transform

$$\frac{dx}{dt} \rightarrow s.X(s)$$

Inductor eqn → $s.\hat{I}_L = -\frac{1-D}{L}\hat{V} + \frac{V}{L}\hat{D}$

Capacitor eqn → $s.\hat{V} = \frac{1-D}{C}\hat{I}_L - \frac{I_L}{C}\hat{D} - \frac{\hat{V}}{RC}$

$$\hat{I}_L = -\frac{1-D}{Ls}\hat{V} + \frac{V}{Ls}\hat{D} \quad -①$$

Substitute ① in capacitor eqn

$$s.\hat{V} = \frac{1-D}{C}\left(-\left(\frac{1-D}{Ls}\right)\hat{V} + \frac{V}{Ls}\hat{D}\right) - \frac{I_L}{C}\hat{D} - \frac{\hat{V}}{RC}$$

$$s.\hat{V} = -\frac{(1-D)^2}{LCs}\hat{V} + \frac{(1-D)V}{LCs}\hat{D} - \frac{I_L}{C}\hat{D} - \frac{\hat{V}}{RC}$$

$$s\hat{V} + \frac{\hat{V}}{RC} + \frac{(1-D)^2\hat{V}}{Ls} = \frac{(1-D)V}{Ls} - \frac{I_L}{C}\hat{D}$$

$$\left(s + \frac{1}{RC} + \frac{(1-D)^2}{Ls}\right)\hat{V} = \left(\frac{(1-D)V}{Ls} - \frac{I_L}{C}\right)\hat{D}$$

$$\left(s^2 + \frac{s}{RC} + \frac{(1-D)^2}{LC}\right)\hat{V} = \left(\frac{(1-D)V}{LC} - \frac{I_L}{C}s\right)\hat{D}$$

final Transfer Eqn $\Rightarrow G_{vd}(s) = \frac{\hat{V}(s)}{\hat{D}(s)}$

$$\boxed{G_{vd}(s) = \frac{\frac{(1-D)V}{LC} - \frac{I_L}{C}s}{s^2 + \frac{s}{RC} + \frac{(1-D)^2}{LC}}}$$

step-3

Substitute $V_g = 12V$, $V_o = 20V$, $L = 74.6\times10^{-6}H$, $R = 4\Omega$
$D = 0.4$, $C = 400\times10^{-6}F$, $I_L = 8.33A$

$$G_{vd}(s) = \frac{\frac{(1-0.4)20}{74.6\times10^{-6}} - \frac{8.33\times s}{400\times10^{-6}}}{s^2 + \frac{s}{4\times400\times10^{-6}} + \frac{(1-0.4)^2}{74.6\times10^{-6}\times400\times10^{-6}}}$$

$$\boxed{G_{vd}(s) = \frac{4.021\times10^8 - 20825s}{s^2 + 625s + 1.206\times10^7}} \quad (eq1)$$

$$\boxed{G_{vd}(s) = \frac{-20825s + 4.021\times10^8}{s^2 + 625s + 1.206\times10^7}}$$

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Step-4

for boost converter → $V_{out} = \frac{V_{in}}{1-D}$

$$R_L = \frac{V_g}{R(1-D)^2}$$

$$G_{vd}(s) = \frac{(1-D)V - IL_s}{LCs^2 + \frac{Ls}{R} + (1-D)^2} \rightarrow \text{substitute}$$

$$= (1-D)^2 \left[\frac{LC}{(1-D)^2} s^2 + \frac{L}{R(1-D)^2} s + 1 \right]$$

Erickson defines $\Rightarrow \omega_0 = \frac{1-D}{\sqrt{LC}}$

$$Q \Rightarrow Q = R(1-D)\sqrt{C/L}$$

Define the RHP zero → $1 - \frac{s}{\omega_z}$

$$\boxed{\omega_z = \frac{R(1-D)^2}{L}}$$

Final canonical form ⇒ $$\boxed{G_{vd}(s) = \frac{V_g}{(1-D)} \cdot \frac{1 - s/\omega_z}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}}$$

Step-5 Substitute values

$V_{in} = 12V$, $V_o = 20V$, $D = 0.4$, $D' = 0.6$, $L = 74.6\mu H$

$C = 400\mu F$, $R = 4\Omega$

$$\Rightarrow G_{do} = \frac{V}{1-D} = \frac{20}{0.6} = 33.33$$

$$\Rightarrow \omega_o = \frac{D'}{\sqrt{LC}} = \frac{0.6}{1.727 \times 10^{-4}} \Rightarrow \boxed{\omega_o = 3474 \text{ rad/s}}$$

$$\boxed{f_o = 553 \text{ Hz}}$$

$$\Rightarrow Q = D'R\sqrt{\frac{C}{L}} \Rightarrow 0.6 \times \sqrt{5.362} \times 4 \Rightarrow \boxed{Q = 5.56}$$

$$\Rightarrow \omega_z = \frac{D'^2 R}{L} = \frac{0.6^2 \times 4}{74.6 \times 10^{-6}} \Rightarrow \boxed{\omega_z = 19303 \text{ rad/s}}$$

$$\boxed{f_z = 3072 \text{ Hz}}$$

$$\boxed{G_{vd}(s) = \frac{33.33 - \frac{33.33\,s}{19303}}{1 + \frac{s}{19315} + \left(\frac{s}{3474}\right)^2}}$$

$$\boxed{G_{vd}(s) = \frac{33.33 - 0.001727\,s}{s^2 + 625s + 1.206 \times 10^7}}$$

Poles $\Rightarrow$ $s_1 = -312.5 + j3460$

$s_2 = -312.5 - j3460$

Zeroes $\Rightarrow +19303$

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Step 6 Select the crossover frequency

$$f_c \leq \left(\min\left(\frac{f_0}{2}, \frac{f_z}{5}\right)\right.$$

$f_c \geq 200 - 250\ Hz$

$f_c = 250\ Hz$

$G_{vd}(jw_c) \Rightarrow w_c = 2\pi f_c$

$\boxed{w_c = 1571\ rad/s}$

Step 6

take cross over freq $f_c = 300\ Hz$

$w_c = 1885\ rad/s$

Step-7 Evaluate the plant at crossover freq

$$s = jw_c = j1885$$

$$\frac{w_c}{w_0} = \frac{1885}{3474} = 0.543$$

$$\frac{w_c}{w_z} = \frac{1885}{19303} = 0.0976$$

$$G_{vd}(s) = \frac{1 - \frac{s}{19303}}{1 + \frac{s}{5.56 \times 3474} + \left(\frac{s}{3474}\right)^2}$$

$$\text{numerator} \Rightarrow 1 - j\frac{w_c}{w_z}$$

$\boxed{\text{magnitude} \Rightarrow |N| = 1.0047}$

$\boxed{\text{phase} \Rightarrow \theta_N = -5.58}$

denominator ⇒ $0.705 + j0.0977$

magnitude ⇒

$$|D| = 0.712$$
$$\theta_D = 7.89°$$

$$|G_{vd}(jw_c)| = 3.33 \times \frac{1.0047}{0.712} = 47.0$$

↓

Plant magnitude

Plant phase ⇒ $\theta = \theta_N - \theta_D$

$$\theta = -13.47°$$

$$|G_c(jw_c)| \times |G_{vd}(jw_c)| = 1$$

↓ 47

$$|G_c(jw_c)| = 0.0213$$

Step 8 taking Desired phase Margin as 60°
(LC usually designed at this)

loop phase = $+60 - 180° = -120°$

Plant phase = $-13.47°$

∴ controller phase = $-120 - (-13.47)$
$= -106.5°$

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step 8 PID Transfer function

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

$$G_c(s) = K_p\left(1 + \frac{\omega_i}{s} + \frac{s}{\omega_d}\right)$$

where, $\omega_i = \frac{K_i}{K_p}$

step 9 rule is to place PI zero at about one decade lower the crossover frequency.

$f_c = 300\,Hz \rightarrow$ one decade lower is $30\,Hz$

$$\boxed{\omega_i = 2\pi(30) = 188.5\,rad/s}$$

step 10 Calculate K_p, K_i, K_d

$$|G_c| = K_p\sqrt{1 + \left(\frac{\omega_i}{\omega_c}\right)^2}$$

$|G_c| = 0.0213 \rightarrow$ from early eqn

$$K_p = \frac{0.0213}{\sqrt{1 + 0.1^2}}$$

$$\boxed{K_p = 0.0212}$$

$$\boxed{K_i = K_p \omega_i}$$

$$\boxed{K_i = 3.99}$$

common initial value of $\boxed{K_d = 3 \times 10^{-6}}$

Step 9 - Controller Transfer function

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

$$G_c(s) = 0.0212 + \frac{3.99}{s} + 3\times10^{-6} s$$

$$\boxed{G_c(s) = \frac{3\times10^{-6}s^2 + 0.02125 + 3.99}{s}}$$

Pole $\Rightarrow$ $\boxed{s = 0}$

$\therefore$ This pole is integrator pole
it removes steady state error

Zeroes $\Rightarrow$ $s = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$s = \frac{0.0212 \pm \sqrt{0.0212^2 - 4(3\times10^{-6})(3.99)}}{2\times3\times10^{-6}}$$

$\therefore$ &

$\boxed{z_1 = -193 \text{ rad/s}} \rightarrow 30.7 \text{ Hz}$

$\boxed{z_2 = -6873 \text{ rad/s}} \rightarrow 1094 \text{ Hz}$